MAE1001

Fall 2026

Screwdriver Disassembly

Presented by Preethi Siva Kumar and Dr. Kartik Bulusu

Course webpage: https://gwu-mae1001.github.io/fall2026/

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Learning Objective

Reverse Engineering

Take apart a screwdriver and identify how it works

Core Engineering Concepts

What are gears and motors, how does it create power (torque)

Ethical AI

Apply ethical AI to understand concepts and Equation.

Core Engineering Concepts I

Torque

Torque is rotational force

τ = *Force x distance*

Power

Power is the rate at which work is done or energy is transferred

Power = Voltage x Current

Efficiency

How efficient the power is used, Output vs Input power

Efficiency = (Output Power / Input Power) x 100

Gear Tooth

How many tooths are in the gear

n = Tooth Number

Gear Ratio

The relation of two meshing gears relative to Gear Teeth number.

Gear Ratio = Driven / Drive

Determines tradeoff between speed & torque

High gear ratio = more torque, less speed

Core Engineering Concepts II

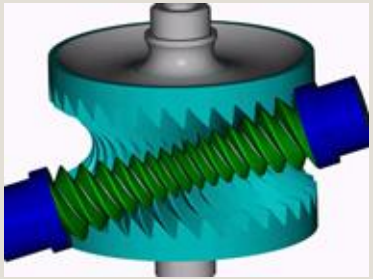
Types of Gears:

Spur Gear



Simple, cost-effective gears with straight teeth that transmit motion and power between two parallel shafts

Worm Gear



Screw-shaped gear (the worm) rotates and meshes with a toothed wheel (the worm wheel) at a 90 deg

Planetary Gear



Sun Gear in the Middle, Multiple Planetary Gears revolving around the Sun Gear, Ring Gear at the edge

Key Concepts

Discuss: How do the gears change torque vs. speed? Why are planetary gears preferred in screwdrivers?

1

DC Motor

Converts electrical energy to mechanical rotational energy

2

Planetary Gears

Multiplies torque while reducing speed and the opposite

3

Trigger Mech

Controls power input

4

Battery & Circuit Board

Supply and regulate electricity

Lab Instructions

Questions will be asked for the Exit Ticket

Materials Needed

Screwdriver per 2 students

Phone to take pictures of steps

Laptop to create lab notebook to record lab

For each step: Add Picture, Add Observations

Lab Instructions

Step 1

Remove Battery Pack



Step 2

Remove Security Clip and take the two sections apart



Step 3

Open the orange casing to reveal the motor



Step 4

Remove all components from inside head to reveal gears



Goal

Now put it back together



<- Motor Gear

Gear ->



1. How many Gear Teeth are on the Motor Gear and the Gear?
2. What is the Gear Ratio between the Motor Gear and Gear?
3. Use AI to Answer the following (Show Prompt and Answer):
    a. Solve the Ring Gear Fixed **Gear Ratio** given the following teeth number, Sun Gear 10, Planet 20, Ring 50

ANS: 6:1

Questions

Add to Lab Notebook

Deliverables

11/21	**Lab**	We're here!	Complete the lab
12/04	**Lab Notebook**		Submit a lab notebook that shows: Each step taken, image of step, and observations and description
12/04	**Ethical AI**		Answer the questions about Gears using AI

Come to Office Hours

Preethi Siva Kumar

Email: preethis@gwu.edu

Any questions? Ask away!

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